Secure and Scalable LAN Design for UrbanBox Fitness Studio

Description:

UrbanBox is a busy city-centre boxing and fitness studio, popular with local residents and office workers. The business offers a range of instructor-led classes, personal training sessions, self-service kiosks for booking classes, staff offices and digital fitness tracking via connected equipment. UrbanBox is opening a new flagship studio and requires a high-performing, secure and reliable wired network for its day-to-day operations.

You have been asked to design and prototype a local area network (LAN) that will support all essential business functions, ensuring efficient administration, secure data handling and consistent member experiences. Your design will be presented to UrbanBox's management team for consideration.

Project Requirements:

Requirement 1: Departmental Segmentation Using VLANs
The LAN must support multiple departments, each segmented into its own VLAN for traffic isolation, security and management efficiency. The VLANs should be set up to optimise network performance while keeping communication secure. The department/user group that need to be separated using different Data VLANs are:

- Reception
- Finance
- Executive
- Instructors
- Members

Consider and include other VLAN types as you deem necessary.

Configure trunk lines between switches and routing devices to support VLAN traffic. To enhance security:

- Allow only the required VLANs on each trunk ("allowed vlan" command).
- Set the native VLAN to a dedicated value other than VLAN 1.
- Disable Dynamic Trunking Protocol (DTP) on all trunk ports.

These measures will help protect against misconfigurations and certain network attacks.

Requirement 2: Inter-VLAN Communication and External Connectivity
The different VLANs must be able to communicate with one another where necessary.

Implement inter-VLAN routing using a method appropriate to your network design. You may use either router-on-a-stick or Layer 3 switching. Clearly state and justify your choice in your report, with reference to scalability, device capabilities and business requirements.

All users should have access to devices and services outside of the gym's network, such as cloud-based fitness applications or internet resources.

Requirement 3: IPv4 Addressing with Dynamic Allocation
Each VLAN should be assigned a unique /24 IPv4 address range. For best practice and clarity, the network address you choose should relate clearly to the VLAN number (for example, VLAN 10 could use 192.168.10.0/24, VLAN 20 could use 192.168.20.0/24). You can use VLAN numbers of your choice but ensure the network address allocation matches.

IP addresses should be allocated dynamically through a DHCP server for efficient network management and ease of use. Ensure all DHCP pools align with your chosen IP IPv4 address ranges.

Requirement 4: Network Redundancy for High Availability
The network must be designed with redundancy to ensure minimal downtime.

Implement redundancy at Layer 2 levels to prevent failures and ensure continuous operation of the network services.

Also consider layer 3 redundancy to allow continued Inter-VLAN communication when the primary routing devices is unavailable.

Requirement 5: Network Device Security and Hardening
Network devices (such as switches and routers) must be hardened to enhance the security of the network and prevent unauthorised access. Securing the network devices will help ensure the availability of services and reduce the risk of attacks.

NOTE: As this is a simulated prototype network, it is recommended to add password once all other requirements have been achieved.

Tasks:

1. **Create a Prototype Network Using Cisco Packet Tracer**
    - Build a network prototype that includes all the specified features, including VLAN segmentation, inter-VLAN communication, IP addressing, redundancy and security hardening.
2. **Report on LAN Design:**
    - Provide a detailed report explaining how the network meets the gym's requirements. The report should describe:
        - How the VLANs are set up to separate traffic between departments.
        - How inter-VLAN communication is achieved and how external connectivity is ensured.
        - The IP addressing scheme, with details on dynamic allocation using DHCP.
        - How network redundancy is implemented to ensure high availability.
        - Security measures taken to harden network devices and improve the security posture.
    - Include screenshots from Cisco Packet Tracer and configuration details to support your explanations.

Submission Guidelines:

- **Due Date:** Saturday, 22nd Nov 2025 @ 9pm
- **Deliverables:**
  - **Prototype**: Cisco Packet Tracer file (.pkt)
  - **Report**: PDF document explaining the design (.pdf)
- **Submission:** Email both ***files*** to *keara.barrett@setu.ie*

Project Interviews/Defence:

- Project defence will commence during the week starting **24th November 2025**. You will be required to demonstrate and explain your prototype to show how it fulfils the gym's network needs. Be ready to discuss your design choices.

Weighting:

- This project accounts for 40% of the 'Networking: Switching and VLAN Concepts' module.